# How Kalman Filters Track Noisy Motion

Speaker manuscript

SKILL Application Test

## Slide 1 How Kalman Filters Track Noisy Motion

Today we will follow a moving object using a sensor that reports noisy positions. The useful output is a trajectory, a velocity estimate, and an uncertainty estimate that changes as information arrives. We will build the estimator from a small numerical example, connect that example to the matrix equations, and then test the result on a reproducible tracking problem.

The central operation repeats at every timestamp. A motion model predicts the next state. A measurement supplies a discrepancy from that prediction. The filter uses the uncertainties of both sources to determine the correction. We will also examine missing measurements, a large injected error, and repeated trials that check whether the reported uncertainty has the expected meaning.

## Slide 2 Track the motion behind noisy measurements

The blue curve is the true position in a constructed motion experiment. The points are the measurements available to the estimator. We have 100 samples separated by 1 second. The position sensor adds independent Gaussian noise with a standard deviation of 3 meters. The moving object also experiences random acceleration, so the curve changes direction and speed over time.

We keep the true trajectory because it lets us measure estimation error directly. The estimator itself receives the position measurements and the specified motion model. Its tasks are to estimate position, infer velocity, and report a covariance that describes the remaining uncertainty. Read the figure as a distinction between the physical quantity we want and the measurements we actually receive. Later figures will use exactly this trajectory to compare parameter choices.

## Slide 3 Represent the state and the measurement separately

The state vector contains position p and velocity v. Position is measured in meters and velocity in meters per second. A hat over a variable denotes an estimate. The sample index k tells us which timestamp we are discussing. Keeping the physical units explicit will help us interpret both covariance entries and gain components.

The sensor observes position, so its measurement matrix H is the row vector containing 1 and 0. Multiplying H by the state selects the position component. Sensor noise is then added to produce z. Its variance R is 9 square meters, the square of the 3-meter standard deviation. Velocity becomes observable through changes in position across

time. The bottom diagram shows the state-to-measurement mapping; the motion model will connect successive states.

### Slide 4 A motion model carries information into the next step

The deterministic prediction uses a constant-velocity transition. Over an interval delta t, position advances by velocity multiplied by elapsed time, while the predicted velocity stays the same. The transition matrix F encodes those relationships. Random acceleration enters through G, with a position contribution of delta t squared divided by 2 and a velocity contribution of delta t.

We describe that acceleration with variance q. In this discrete model, its covariance contribution to the state is q times G G transpose. At a 1-second interval and q equal to 0.2, the process covariance Q has entries 0.05, 0.10 and 0.20 as displayed. The cross terms arise because the same acceleration affects position and velocity. Each covariance entry carries units determined by its corresponding state components. Both the dynamics and uncertainty propagation therefore need the actual elapsed time.

### Slide 5 Covariance describes the shape of uncertainty

The covariance matrix summarizes the estimation error in both state components. Its diagonal entries are position and velocity variances. The off-diagonal entry measures how errors in those components vary together. A positive cross-covariance means that a larger position error tends to accompany a larger velocity error.

The contours turn that matrix into geometry. These are joint 95 percent Gaussian contours, drawn on position-error and velocity-error axes. The initial covariance is diagonal, so its principal directions align with the axes. Prediction makes the ellipse tilt because uncertain velocity affects the next position. A position measurement then changes its size and shape. This example starts from diagonal entries 4 and 1, predicts for 1 second with acceleration variance 0.04, and updates using measurement variance 9. The contour equation uses the complete state estimation error x minus x-hat.

### Slide 6 First combine two scalar estimates

Before using matrices, consider a scalar position update. The prior mean is 10 meters with variance 4 square meters. The new measurement is 13 meters with variance 9 square meters. The minus superscript indicates the value before incorporating the measurement, and the plus superscript indicates the value after the update.

The gain is the prior variance divided by the sum of the prior and measurement variances. Here it is 4 divided by 13, approximately 0.308. The discrepancy is 3 meters, so the estimate moves about 0.923 meters toward the observation and becomes 10.923 meters. Its variance becomes 2.769 square meters. On the graph, the posterior combines the prior density with the measurement likelihood. Its location reflects the relative

uncertainty of the sources, and its narrower shape reflects the information gained by combining them.

### Slide 7 Predict both the state and its covariance

The prediction step advances the mean with F and advances the covariance with F P F transpose plus Q. The same model acts on both the estimated trajectory and the uncertainty around it. The added Q accounts for changes that the deterministic prediction does not specify exactly.

For the worked example, start at position 10 meters and velocity 2 meters per second, with covariance diagonal entries 4 and 1. Use a 1-second interval and acceleration variance 0.04. The predicted state is position 12 and velocity 2. The predicted covariance has position variance 5.01, velocity variance 1.04 and cross-covariance 1.02. The component equation on the left explains the position-variance increase: previous position uncertainty, cross-covariance, velocity uncertainty propagated over time, and process noise all contribute. We carry the resulting state and covariance into the measurement update.

### Slide 8 The innovation measures a prediction error

The innovation y is the observed measurement minus the predicted measurement. For a position sensor, the predicted measurement is the predicted position. Its covariance S includes the projected state uncertainty and the sensor variance. This gives the discrepancy a scale that is specific to the current prediction and observation.

Continue the numerical example with a measurement of 13 meters. The predicted position was 12, so the innovation is 1 meter. Its variance is 5.01 plus 9, giving 14.01 square meters. Dividing the innovation by the square root of S gives 0.267. Under the stated Gaussian model, a 95 percent reference interval for the innovation extends to about plus or minus 7.34 meters. The observed 1-meter discrepancy is therefore near the center of the predicted distribution. This normalization will later support both diagnostics and outlier screening.

### Slide 9 One position observation can correct velocity too

The matrix gain projects uncertainty between state space and measurement space. Here H selects position, so P minus times H transpose is the first column of the predicted covariance. Dividing that column by S gives a position gain of about 0.3576 and a velocity gain of about 0.0728 per second.

Multiplying those components by the 1-meter innovation updates the state to position 12.3576 meters and velocity 2.0728 meters per second. The velocity update comes from the positive cross-covariance created during prediction. A position observation that is higher than expected suggests that the object may also be moving faster. The table changes the sensor variance while holding the prediction fixed. Reducing R from 9 to

1 increases both gain components. A more precise sensor has more influence on the corrected state.

### Slide 10 Update uncertainty with the corrected state

Once we change the state estimate, we also update the covariance. The Joseph form uses A equal to I minus K H and combines A P minus A transpose with K R K transpose. These terms make the remaining predicted-state uncertainty and the incorporated measurement-noise contribution explicit.

The numerical covariance on the left is the prediction before the measurement. After the update, position variance is about 3.2184 and velocity variance about 0.9657. Position standard deviation falls from 2.238 meters to 1.794 meters. The off-diagonal covariance changes as well, because the observation affects the joint uncertainty. In code, the Joseph expression is useful for retaining the covariance structure through floating-point calculations. We will also remove tiny numerical asymmetry by averaging the computed covariance with its transpose.

### Slide 11 The estimator is a repeating predict-update loop

Follow the diagram clockwise from the previous estimate. We begin with the corrected state and covariance from the preceding timestamp. Prediction advances both quantities. The new measurement produces an innovation and its variance. Those determine the gain, which corrects the state. The covariance update completes the cycle and supplies the next prediction.

This arrangement also makes operational behavior clear. When a measurement is missing, prediction can still advance to the next timestamp. The predicted covariance grows according to the motion model and process uncertainty. When observations return, the usual correction resumes. The carried state has a fixed size: the state vector and its covariance summarize the measurements already incorporated. Logging the full history is useful for analysis, while the recursive estimator itself only needs its current state and covariance.

### Slide 12 The tracking run reduces position error

Return to the 100-sample tracking run. The left panel plots position error, defined as estimate minus truth. The shaded band shows plus or minus 1.96 times the estimated position standard deviation. The right panel shows true velocity, the inferred velocity, and the much noisier differences of successive position measurements.

For the numerical summary, we evaluate steps 10 through 99 after the initial transient. RMSE means root mean square error: square each error, take the average, then take the square root. The measurement position RMSE is 3.133 meters and the filtered position RMSE is 2.143 meters. The interval is a different output from the trajectory error. It reports the uncertainty predicted by the model; later repeated trials will test

its calibration. Keeping both views helps us assess the estimate and the uncertainty it carries.

### Slide 13 Velocity is inferred through motion over time

A direct velocity estimate divides the difference between consecutive position measurements by the elapsed time. Each difference contains noise from both observations. For independent sensor noise, the variance of this contribution is 2R divided by delta t squared. With R equal to 9 and delta t equal to 1 second, that is 18 square meters per square second.

In the tracking run, the direct-difference velocity RMSE is 4.204 meters per second. The velocity component of the Kalman state has RMSE 0.848 meters per second. The model combines position changes over time with uncertainty information. The small observability matrix at the bottom explains why this can work. H sees position immediately, while H F includes a velocity term multiplied by elapsed time. With nonzero elapsed time, these rows have rank 2, allowing both state components to affect observations across time.

### Slide 14 Process noise controls responsiveness

The process covariance expresses how much motion can change between observations. Here we vary the assumed acceleration variance q while keeping the measurements fixed. The generating process uses q equal to 0.2. The gray estimate uses 0.002, the red estimate uses 0.2 and the green estimate uses 2.

The left plot shows position error and the right plot shows a segment of the velocity trajectory. A smaller q makes the estimate persist more strongly with the model, producing a smoother response that can lag changing motion. A larger q permits more rapid movement in the estimate. The table quantifies this run: position RMSE is 4.401, 2.143 and 2.445 meters for the 3 settings. Velocity RMSE is also lowest at q equal to 0.2. Use such controlled comparisons to connect the tuning parameter to the behavior that matters in the application.

### Slide 15 Read the innovation and gain as diagnostics

The standardized innovation compares each discrepancy with the uncertainty predicted for that comparison. The dashed references are at plus and minus 1.96. This makes the residual sequence easier to interpret over time than raw errors alone. Persistent shifts or sustained changes in its scale can point to a changed sensor or a changed motion regime.

The right panel shows the position and velocity gain components. Their initial movement reflects the initial covariance and the information arriving from the first observations. They then settle as the covariance recursion approaches a stable regime for the repeated model and measurement configuration. These curves make the estimator

inspectable: the innovation describes incoming disagreement, while the gain shows how strongly that disagreement changes the state. The velocity gain has units of inverse seconds, matching the conversion from position discrepancy to velocity correction.

### Slide 16 Missing measurements increase uncertainty

The gray interval marks a gap in observations at steps 40 through 50. During this gap, the filter keeps applying prediction. Its trajectory continues according to the current motion estimate, and the covariance accumulates the effect of uncertainty in velocity and acceleration.

The position standard deviation is 1.944 meters at step 39. By step 50 it has grown to 13.863 meters. When the next observation arrives, the correction reduces it to 2.944 meters. Read the left and right panels together: the first shows the predicted path during the gap, and the second shows how uncertain that path has become. This is why missing observations need an explicit branch in an implementation. The estimator still advances in time, and its uncertainty reflects the information that has not arrived.

### Slide 17 Use the innovation scale to screen a large outlier

For this test, add 30 meters to the observation at step 65. The gray curve accepts every update and is pulled away from the true trajectory. The red curve uses an innovation gate. Its decision is based on normalized innovation squared, or NIS, which is y squared divided by S for this scalar measurement.

Under the stated Gaussian measurement model, NIS has a chi-squared reference distribution with 1 degree of freedom. The 99 percent threshold is 6.635. The injected observation has NIS about 38.42, so the gate skips that correction and keeps the prediction at this timestamp. The following observations resume normal updating. This establishes a concrete handling policy for unusually large discrepancies. The threshold, model assumptions and behavior after a rejected update should all be explicit in the implementation.

### Slide 18 Check covariance calibration across repeated trials

Repeated trials show how often the true position lies inside the reported uncertainty intervals. Here we run 400 independent trials, evaluate 90 steps from each, and compare different assumed sensor variances. The generating model uses q equal to 0.2 and R equal to 9.

The left panel reports mean NIS. With the matched sensor variance, it is 1.007, close to the scalar reference value of 1. The right panel reports interval coverage: the fraction of true positions falling inside the estimated 95 percent position intervals. The matched case reaches 95.1 percent. Assuming R equal to 1 produces narrow, overconfident intervals and coverage of 51.7 percent; assuming 36 produces broader intervals and

coverage of 99.8 percent. These are measurable consequences of the covariance choices. Calibration checks connect reported uncertainty to repeated outcomes.

### Slide 19 Translate the equations into a reusable update

The function starts with prediction for x and P. If z is missing, it returns those predicted quantities. Otherwise it computes the innovation y and its covariance S. The gain is obtained by solving a linear system. This expresses the same gain equation while avoiding the need to form a separate matrix inverse explicitly.

The state update uses K times y, and the covariance update uses Joseph form. The final symmetrization removes small numerical discrepancies between matching off-diagonal entries. Array shapes matter: the state has length 2, H is 1 by 2, the measurement has length 1, and R is a 1 by 1 matrix. The caller builds F and Q for the elapsed time and applies any observation-gating policy. Keeping that contract explicit makes the update usable in a stream of observations with a clear state representation.

### Slide 20 Build the estimator around a testable motion model

The design decisions now form a complete chain. Choose physically meaningful state variables and units. Define how the sensor observes them. Build the transition and process covariance from the time interval and motion assumptions. Set the initial uncertainty, characterize measurement noise, and use observed errors and innovations to assess tuning.

The operating rules also belong in the design. Specify how prediction advances through gaps and how unusually large innovations are handled. Validate the estimated trajectory against known truth when available, and test uncertainty coverage across repeated trials. The main idea to retain is that the filter carries an estimate and a quantitative description of its uncertainty together. Prediction evolves that pair through the model, and measurement updates revise that pair using new information.